Recirculation-Delayed Anechoic Chamber (RDAC) at Oklahoma State University: Design, Construction, and Performance Evaluation

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I. Introduction

An anechoic chamber is a specially designed room that absorbs certain sound waves, creating a reverberation-free environment. This is also known as the acoustical free-field [1]. To maintain acoustic accuracy below a specific cut-off frequency, the chamber's ceilings, walls, and floors must be treated with specially designed acoustical absorption materials. This treatment reduces the anechoic chamber's background sound pressure level (SPL), ensuring minimal reverberation interference. Rusz investigated six different types of acoustic wedges and liners, and the study showed that almost all structures meet free-field requirements at high frequencies [1]. Anechoic chambers offer significant advantages for initial-stage research on small unmanned aerial systems (UAS) and propellers. It enables obtaining reliable data such as acoustic pressure, frequencies, and directivity of noise sources in a controlled indoor environment. Experiments such as evaluating geometric and material improvements to quiet propellers can benefit greatly from a test setup in an anechoic chamber [2]. As the latest study observed substantial differences in SPL and dynamic measurements when comparing indoor and outdoor tests conducted in the same facility, highlighting the importance of controlled acoustic environments [3].

The primary areas of interest for noise prediction for vertical takeoff and landing (VTOL) aircraft or propeller testing are typically in the acoustic far field [4]. The far-field region is defined as "observer distance must be much larger than the acoustic wavelength, depending on the frequency of interest" [5]. This field is preferred because the near field noise sources can be neglected, which explains the commonality for observers in engineering applications to locate the data collection instrument in this region. A practical guideline in propeller testing is to measure the noise at least 10 rotor radii away [6]. This microphone placement will minimize the downwash effects that can distort acoustic measurements and obtain more accurate and representative acoustic data for validating models or simulations. An ideal anechoic chamber should facilitate far-field conditions while simulating a free-field environment to ensure reliable data collection.

Noise from VTOL aircraft consists of various sources, but the most dominant factor that affects community acceptance is the aerodynamic or flow noise [7]. Although the airframe contributes marginally to noise generation, the predominant noise comes from the rotor blades [8]. Blade noise is classified into tonal and broadband noise. Tonal noise consists of thickness noise, caused by the displacement of air by the rotor blade, and loading noise, caused by the force exerted on the fluid by the rotor blade surface [9, 10]. Broadband noise includes airfoil self-noise, blade wake interaction (BWI) noise, and turbulence ingestion noise. Airfoil self-noise is produced by the scattering of turbulent flow over the blade trailing edge [11, 12]. BWI noise occurs when blades interact with wake turbulence, while turbulence ingestion noise stems from unsteady loading noise due to ingestion of atmospheric turbulence into the rotor [13, 14]. Airfoil self-noise can be further categorized into five types. Firstly, the Turbulent Boundary Layer-Trailing Edge (TBL-TE) noise is due to the pressure scattering from the convection of turbulence of the trailing edge. Secondly, the Turbulent Boundary Layer-Separation/Stall (TBL-SS) noise happens at high angles of attack due to the convection of larger-scale turbulent structures past the trailing edge. Thirdly, the Laminar Boundary Layer-Vortex Shedding (LBL-VS) noise is due to vortex shedding past the trailing edge coupled with laminar boundary layer development. Fourth, the Trailing-Edge Bluntness-Vortex Shedding noise is the vortex-shedding noise from a blunt trailing edge. Lastly, the Tip Vortex Formation (TVF) noise is due to the formation and shedding of the blade tip vortex [10–12, 15].

A previous study by Stephenson et al. [16] found that flow recirculation in an enclosed anechoic chamber can affect Unmanned Aerial Vehicle (UAV) acoustic measurements. As a rotor system generates thrust, both tonal and broadband acoustic emissions move into the downwash, and in an enclosed environment, this downwash recirculates within the

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chamber. The recirculated flow is then re-ingested by the rotor system, altering the noise measurement. The study observed that tonal noise measured in-plane increased by over 15dB, while out-of-plane microphone readings showed an increase above 30 dB [16]. Figure 1 shows a propeller that was tested by Lescault et al., a slipstream occurred with a floor or wall that blocked the flow and caused recirculation [17]. Flow recirculation is a function of rotor size, thrust, and the size of the facility, therefore, the occurrence of flow recirculation differs for every facility and usually occurs a few seconds after reaching steady-state RPM [16].

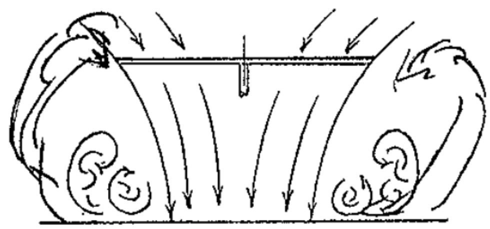

Fig. 1 Illustration of rotor recirculation near a solid surface [17]

As mentioned above, enclosed anechoic chambers are known to experience flow recirculation, which can significantly impact the acoustic and dynamic measurements of propellers or UAV . Examples of such facilities include the NASA Langley Small Hover Anechoic Chamber (SHAC)[16], Mississippi State University's Raspet Flight Research Laboratory (RFRL) [18], Hong Kong University of Science and Technology's Aerodynamic and Acoustic Facility anechoic chamber [19], U.S. Army Aeromechanics Lab's anechoic chamber [17], among others. Given the occurrence of flow recirculation, Weitsman et al. and Wolverton have investigated the use of turbulence suppression screens as a mitigation method and delay the onset of recirculation. The studies show a consistent result of delayed onset recirculation, with dynamic measurements similar to baseline after employing turbulence suppression screens [18, 20]. Although numerous facilities have been constructed to mitigate the recirculation, the mechanisms and the best practices of designing the recirculation-delaying devices are not well understood.

The first objective of this paper is to describe the design, construction, and performance of the enclosed 27 ft x 21 ft x 10 ft anechoic chamber at Oklahoma State University's Rotorcraft Aerodynamic and Aeroacoustics (RAA) Lab in Helmerich Research Center. The second objective is to identify the presence of recirculation and to attain a solution to delay the onset recirculation issues. The third objective is to measure and identify the noise sources from the propeller.

II. Methodology and Experiment Setup

The chamber is designed with a low cut-off frequency of 250Hz to facilitate the rotorcraft and UAV propeller testing. Figure 2 illustrates the design of the anechoic chamber. Acoustic wedges were used for chamber construction, and the recirculation effects will be mitigated by applying various turbulence suppression screens to delay the onset recirculation.

A. Acoustic Wedge

The structure and design of the acoustic wedge were first determined. The Harvard linear wedge is used to construct the RDAC facilities because it is known for its absorption of the acoustical energy in low frequencies. The cutoff frequency is defined as the frequency at which the pressure reflection rises to 10% in a normally incident sound wave. This corresponds to the frequency at which sound energy absorption drops to 99% or at which there is a 20dB reduction for a single reflection [21]. Based on the predetermined lower cutoff frequency of 250 Hz and through Beranek et al. study, the taper length, L_1, should be at least 9.6 inches, while the base length, L_2, should be 2 inches, making the total length, L, approximately 12 inches.

For validation purposes and to ensure effective dissipation of acoustic energy, the following Equation (1) and Equation (2) must be satisfied. λ is the wavelength of the lowest frequency f of interest and L is the length of the wedge[1]. After substituting λ into the dissipation equation, the total length of the wedge should be at least 13.504 inches. Therefore, the design wedge length of RDAC is determined to be 14 inches.

$$\frac{L}{\lambda} \geq 0.25 \tag{1}$$

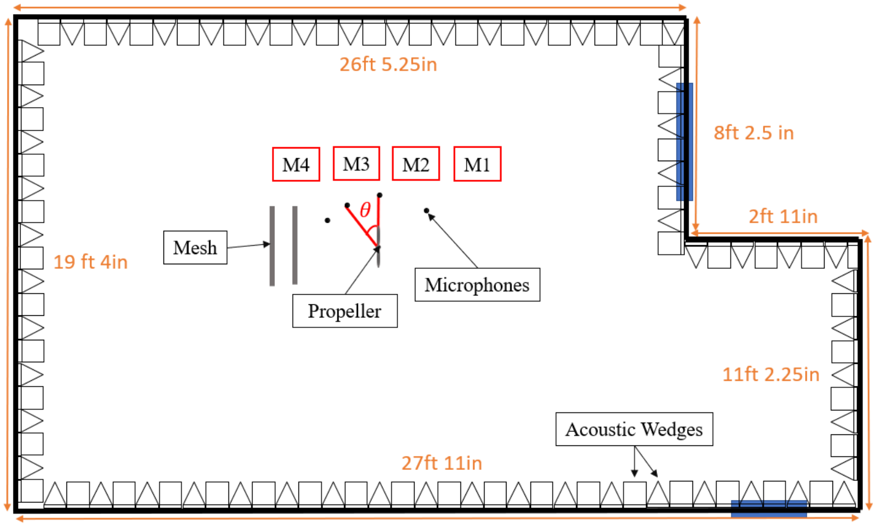

Fig. 2 Recirculation Delayed Anechoic Chamber (RDAC) Envisioned Top View

$$\lambda = \frac{c_0}{f} \tag{2}$$

Regarding wedge material, melamine was used due to its fire resistance [22]. In addition, melamine material was used because the foam possesses a fully open three-dimensional mesh structure. With the appropriate length, it enables sound waves to penetrate the foam's deep layers, where the waves rapidly dissipate through the vibration of the melamine foam's structure[23]. As a result, reflected sound waves are effectively diminished. After iterative design considerations using the equations and a thorough literature review, the Harvard Linear Wedge used in constructing the anechoic chamber in RAA lab is determined to have a base length, L_2, of 2 inches, and a total length, L, of 14 inches, utilizing melamine materials. Figure 3 presents the CAD model of the desired Harvard Linear Wedge. The acoustic wedges were then installed in an alternating manner, 90 °perpendicular to one another to minimize reverberation, especially for frequencies beyond the designated 250 cutoff frequency. This ensured the chamber has maximum wedge coverage while maintaining precise horizontal and vertical alignment. This step was critical to achieve uniform acoustic absorption and a consistent layout across all chamber surfaces. The completed anechoic chamber contained approximately 2,054 acoustic wedges.

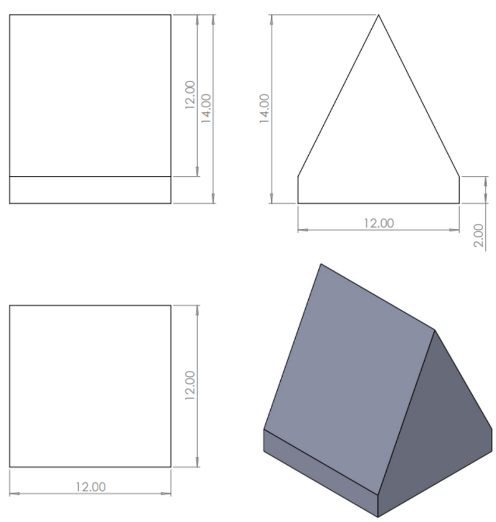

Fig. 3 CAD Model of the Acoustic Wedge Used in RDAC

B. Mesh Screen

Turbulence suppression has been extensively studied, with mesh screens and honeycomb structures commonly used in wind tunnels. When turbulent flow passes through a mesh screen, it behaves as if it were flowing through a parallel interconnecting channel with varying cross sections [24]. In wind tunnels, mesh screens function as flow straighteners, reducing velocity lateral components and minimizing boundary layer effects caused by turbulence. However, with the increasing use of enclosed anechoic chambers, issues such as flow recirculation within chambers emerge. As a result, turbulence suppression mesh screens are now commonly employed in anechoic chambers.

Weistman et al. studied several configurations of mesh screen placement and their effects on mitigating flow recirculation. Their study found that a dual-mesh downstream of the propeller resulted in the longest delay onset recirculation after reaching steady-state RPM [20]. In addition, many studies regarding mesh have focused on its correlation with pressure drop. Given a total pressure drop, it is advantageous to use several screens with a relatively low-pressure drop coefficient. A study conducted by Groth et al. found that the turbulence-damping ability of a screen increases as mesh grid size decreases. In addition, when multiple screens are used in combination, their separation should be larger than the length of the initial decay region. However, using multiple fine mesh screens only in combinations results in a significant pressure drop, which is not ideal. Therefore, a combination of a relatively coarse screen followed by a finer one is the most effective at reducing turbulence while minimizing pressure drop [25]. In this study, mesh screens are positioned downstream of the propeller shown in fig. 2. The larger grid size mesh screen will be placed closest to the propeller, with the smaller mesh grid placed downstream of the first screen. Various mesh grid opening sizes and various combinations of downstream mesh screens will be tested to analyze these factors' effects on recirculation.

C. Testing Instruments

Acoustic data in this study were recorded in the time domain as sound pressure signals using four 1/2-inch Brüel & Kjaer precision microphones (Type 4966-H-041) connected to a National Instruments NI-9234 data acquisition module. NI FlexLogger was calibrated to 46.5 mV/Pa for microphone sensitivity and used for data acquisition and real-time signal visualization. The time-domain signals were processed using the Fast Fourier Transform (FFT) algorithm within NI FlexLogger, converting the pressure-time signal into frequency-domain spectra. This yielded the data presented in SPL (dB) as a function of frequency (Hz), allowing for comprehensive frequency-based acoustic analysis and direct comparison between test conditions.

A single 15 x 5 Carbon Fiber T-motor fixed-pitch propeller was mounted on a T-motor MN2212-21 motor. The motor RPM was controlled using an electronic speed controller (ESC) connected to RC Benchmark to ensure steady and repeatable rotational speeds during each test. The data were recorded once the propeller achieved steady-state RPM conditions to maintain consistency across multiple trials. Figure 4 below shows the initial experimental setup, and table 1 shows the locations of the microphones.

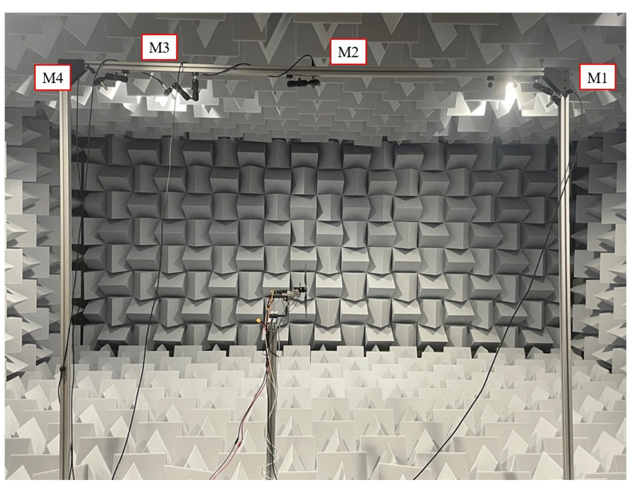

Fig. 4 Motor-Rotor Test Setup with Microphone Locations

Microphone	M1	M2	M3	M4
Distance (in)	56.2	41.0	47.0	57.8
Elevation Angle (θ)	43.67	0	-29.99	-44.14

Table 1 Microphone Placement

III. Preliminary Results

The propeller noise was measured with the distance of the microphone greater than five times the radius of the propeller, 37.5 in. This ensures that the testing performed is in the far-field region. The propeller was tested in three rotational speeds: 3720 RPM, 5190 RPM, and 7150 RPM. The SPL data collected from all four microphones were averaged for each RPM to examine the acoustic signature at low, mid, and high rotational speeds in the anechoic chamber. Figure 5 below shows the SPL across all three RPM levels with respect to a logarithmic frequency axis.

It is worth noting the presence of three unexpected SPL spikes at 982 Hz, 1039 Hz, and 1098 Hz during the 5190 RPM test. The 5190 RPM test was run immediately after the 7150 RPM, this may have influenced the airflow patterns inside the chamber. This led to the thought that these spikes may be due to the recirculation effects within the chamber, which is a common concern in enclosed testing environments. This problem will be further investigated in the full paper.

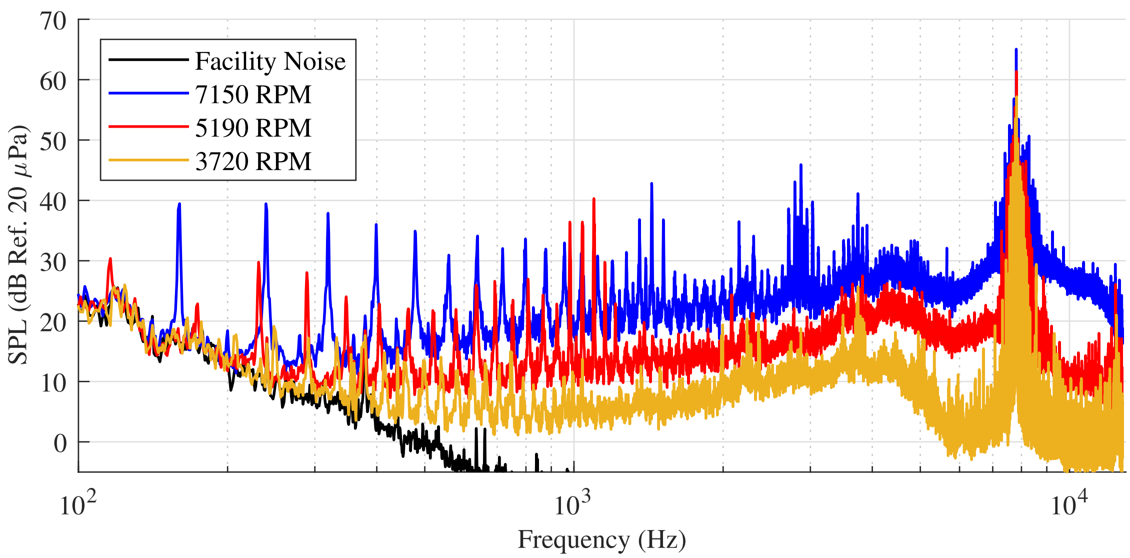

Fig. 5 Measured Frequency Spectra of a two-bladed 15-in Propeller in RDAC at 3720 RPM, 5190 RPM, and 7150 RPM

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A. Propeller Noise Directivity

As for the directivity test, microphone M1 was located upstream of the propeller at a 43.67 °. M2 was in-plane with the propeller, 0 °. M3 was located downstream of the propeller at a -29.99 °, and M4 was located -44.14 °downstream. The data collected were used to study the SPL trend at each microphone location. Each RPM condition served as a verification of the consistency of the noise directivity pattern.

The SPL frequency spectra of a two-bladed 15-in propeller at 3720 RPM, 5190 RPM, and 7150RPM for each microphone locations M1, M2, M3, and M4 can be seen in fig. 6, fig. 7, and fig. 8 below. When further analyzed using Overall Sound Pressure Level (OASPL), the M2 location exhibited the lowest OASPL across all RPMs. This is consistent with typical propeller polar plots, where the noise signature often dips at in-plane. This confirms the bi-directional nature of propeller noise. While M1, M3, and M4 do not show a specific trend, they consistently have higher OASPL values than in-plane microphones and vary less than 1 dB among themselves. Table 2 below summarizes the OASPL of each microphone location at each RPM.

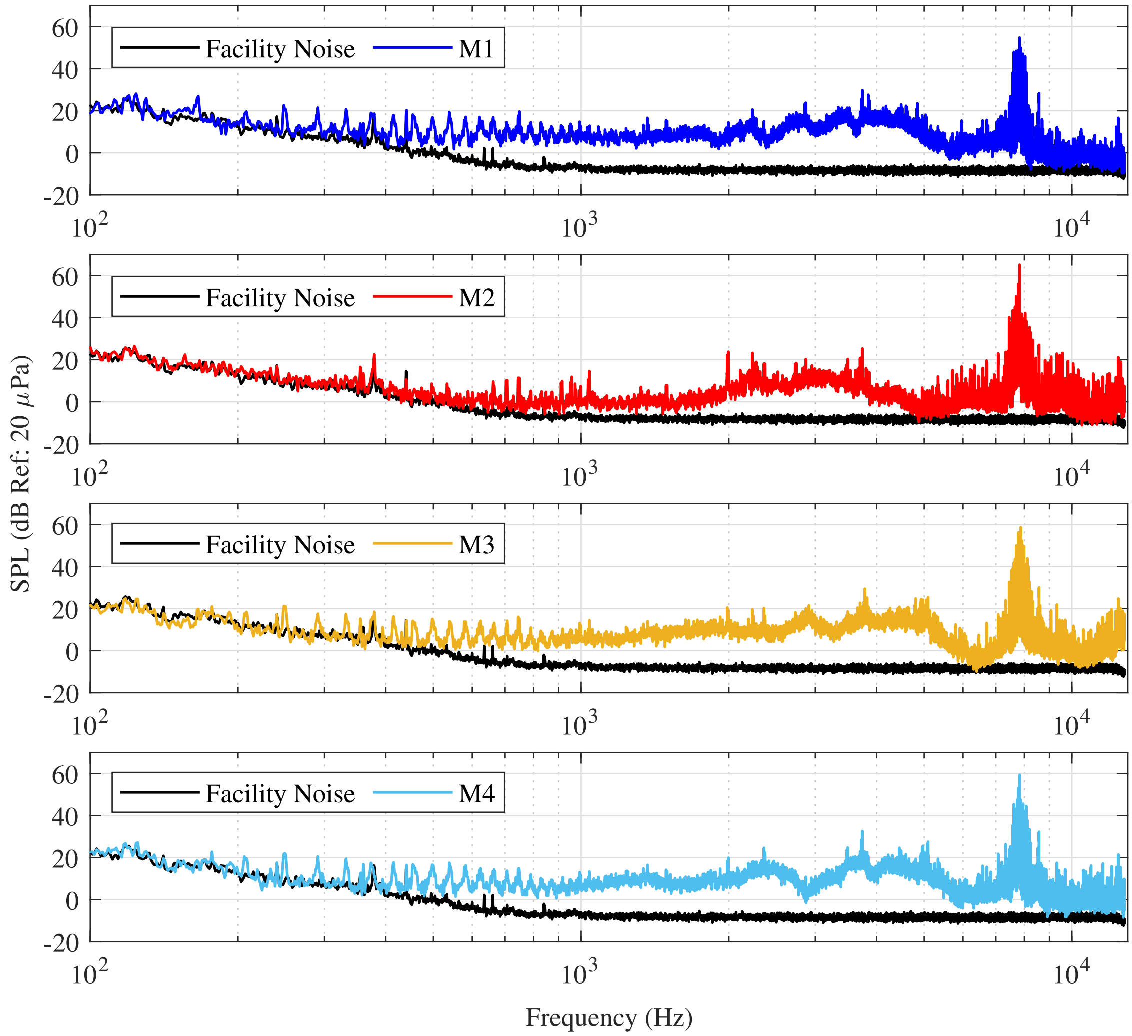

Fig. 6 Measured Frequency Spectra at 3720 RPM at Microphone Locations M1, M2, M3, and M4 of a two-bladed 15-in Propeller in RDAC

RPM	OASPL (dB Ref. 20 μPa)			
	M1	M2	M3	M4
3720	95.45	94.72	95.87	95.94
5190	102.88	99.36	102.43	103.10
7150	111.96	106.84	111.16	111.57

Table 2 Overall Sound Pressure Level (OASPL) Measurements at Various RPMs

IV. Full Paper Plan

The current preliminary results only satisfy the first objective of the paper, which is to design, construct, and evaluate the performance of an enclosed anechoic chamber. The next steps are to complete the second objective of the paper, which is to identify recirculation in the anechoic chamber and to study further how to delay recirculation using turbulence suppression mesh screens. After completing that, the team plans to finish the paper by measuring and identifying the noise sources from the propeller in the RDAC, which will satisfy the third objective.

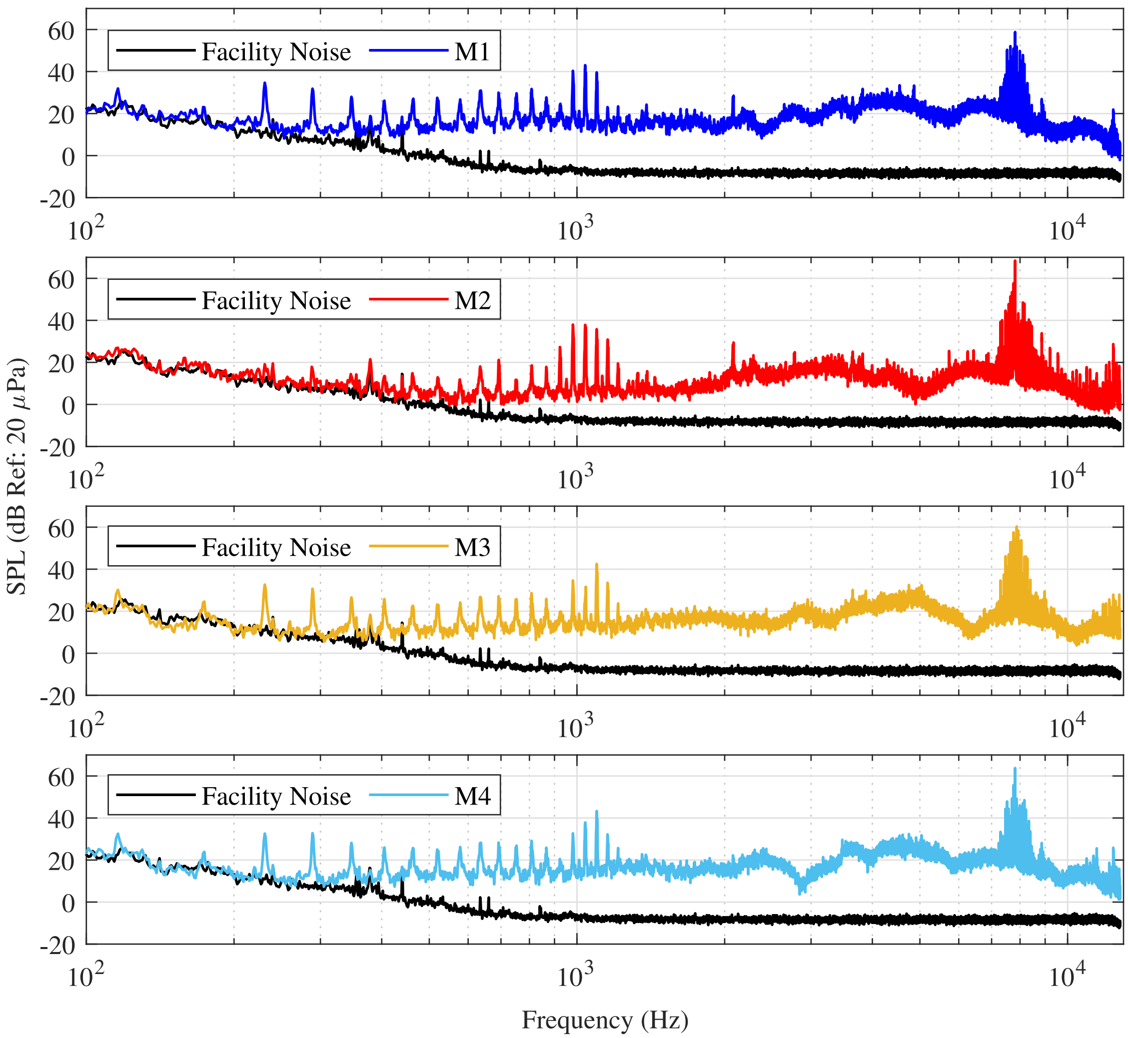

Fig. 7 Measured Frequency Spectra at 5190 RPM at Microphone Locations M1, M2, M3, and M4 of a two-bladed 15-in Propeller in RDAC

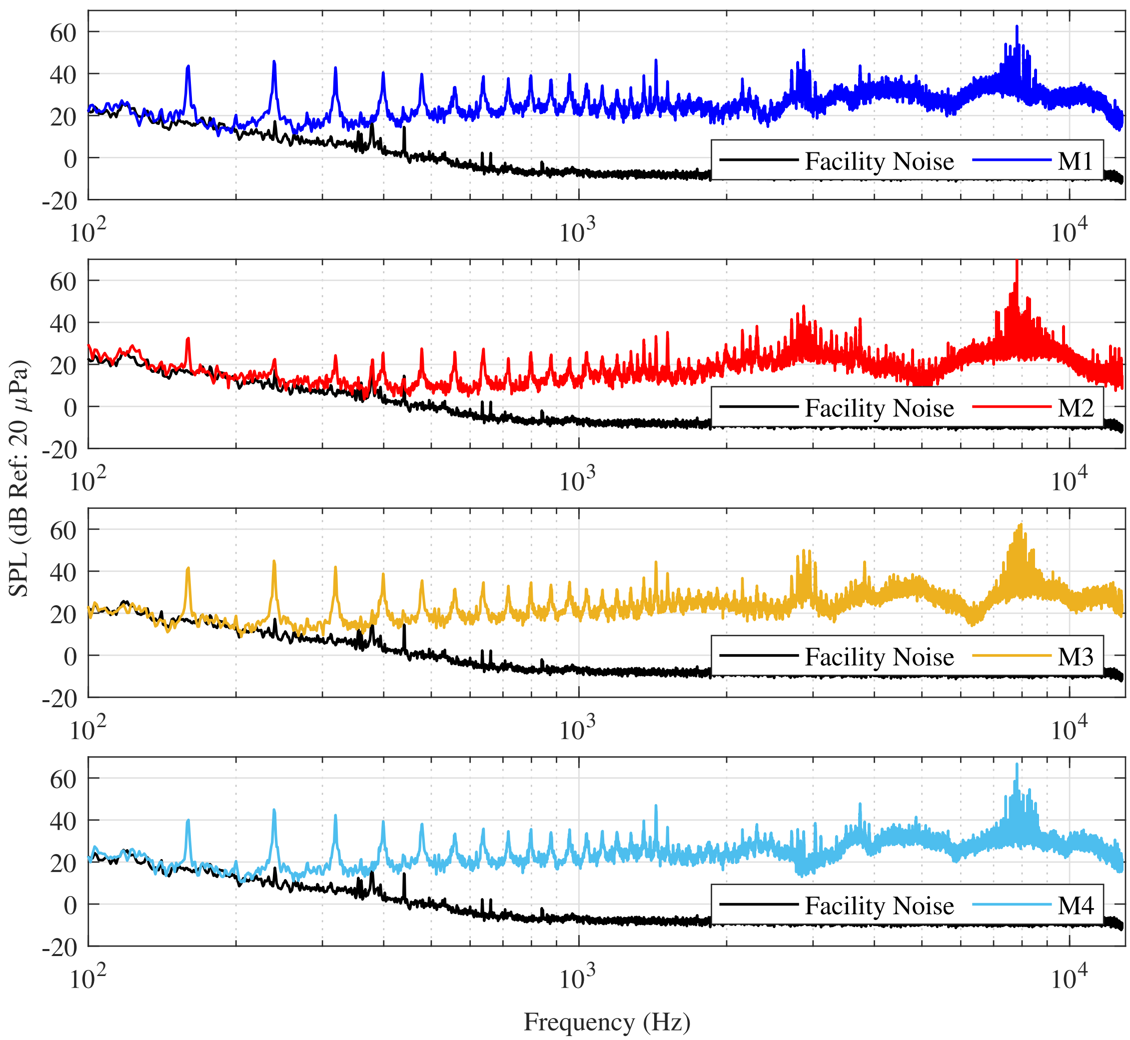

Fig. 8 Measured Frequency Spectra at 7150 RPM at Microphone Locations M1, M2, M3, and M4 of a two-bladed 15-in Propeller in RDAC

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